State Is the Interface: Evaluating Handoff Fidelity in Multi-Agent LLM Workflows

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Abstract

Multi-agent language-model systems increasingly transfer task ownership between specialized agents, yet existing evaluations rarely measure which task-critical state survives the transfer. We introduce HandoffBench, a controlled evaluation protocol for inter-role handoffs in policy-constrained tool workflows. Each episode fixes an upstream trace and annotates hidden boundary state: verified evidence, epistemic unknowns, scoped consent, and policy decisions. Deterministic receiver-state probes link field fidelity to regression and policy violations. We compare full history, free-form summaries, structured payloads, and Executable Handoff Capsules, which bind typed claims to trace provenance and machine-checkable preconditions. We seal a pre-execution factorial analysis plan to isolate typing, provenance, and executable checks under matched evidence and budgets. Before candidate-model evaluation, isolated agents double-annotated and audited, replaced exclusions, and sealed a balanced 200-family split. Across 8,800 intent-to-treat runs with two models, two seeds, and eleven conditions, Structured transfer achieved 71.6% strict success versus 91.1% for the Gold Oracle, a prespecified gap of 19.5 percentage points (family-clustered 95% CI $[15.4, 23.6]$; Holm-adjusted $p < .001$). Pooled advisory checks improved strict success by 3.63 points (95% CI $[1.91, 5.44]$; Holm-adjusted $p < .001$). Exploratory estimates were $+7.06$ points for Ministral and $+0.19$ for Qwen, revealing model-dependent check utility. Substantial boundary regression thus remains even with structured state.

1 Introduction

Handoffs are becoming a standard orchestration primitive: an intake agent may collect constraints, a policy agent may verify eligibility, and a resolution agent may execute the final action. The routing decision can be correct while the workflow still fails. A verified fare can disappear, an unresolved field can become an assumed value, or approval for one action can be mistaken for consent to another. These are failures of the interface between agents.

Existing agent benchmarks richly measure tool use, policy compliance, and team performance (Liu et al. 2024; Yao et al. 2024; Lu et al. 2025; Zhu et al. 2025), while recent coding-agent takeover evaluation measures the effort needed to resume interrupted work (KC and Budathoki 2026). These works do not, however, expose a gold semantic state at an inter-role boundary and score which atomic facts, unknowns, permissions, and preconditions crossed it. Consequently, their end-to-end scores do not directly quantify regression attributable to a changed boundary representation under a fixed receiver and environment.

We present HandoffBench, a benchmark and intervention protocol that holds the upstream trace, target model, tools, and budget fixed while intervening on the handoff representation. Its central object is a task-critical annotated gold state whose claims are typed and linked to authenticated user, tool, policy, or environment events. Core metrics are computed from typed equality, provenance integrity, simulator transitions, and Boolean success predicates rather than an LLM judge.

Our contributions are: (1) a controlled benchmark for field-level handoff fidelity in interactive tool workflows; (2) deterministic state probes and oracle interventions that identify handoff-induced regression; and (3) a matched factorial protocol that separates typing, provenance, and advisory executable checks from action enforcement; and (4) a pre-execution-sealed 8,800-run evaluation showing a 19.5-point Structured-to-Oracle boundary gap and a 3.63-point average benefit from advisory checks. Secondary estimates for typing and provenance are reported separately from the confirmatory multiplicity family.

2 Method

Boundary-state evaluation. HandoffBench isolates a single transfer of control rather than evaluating generic conversational memory. An episode is $e = (E, \mathcal{A}, \mathcal{U}, \tau, b, a_s, a_t, G_b, \Phi)$, where E is a seeded workflow environment, τ is the authenticated source trace cut at boundary b, a_s and a_t are the source and receiver roles, G_b is evaluator-private boundary state, and Φ is a machine-executable terminal predicate. For transfer mechanism m, the receiver observes only $V_m(\tau)$ and produces post-boundary trace ρ_m. Across mechanisms we hold fixed E, τ, b, receiver model and prompt, tools, decoding parameters, and seed. Thus, comparisons vary the transfer representation while preserving the evidence and downstream capability. Figure 1 summarizes the controlled path and evaluator-only oracle

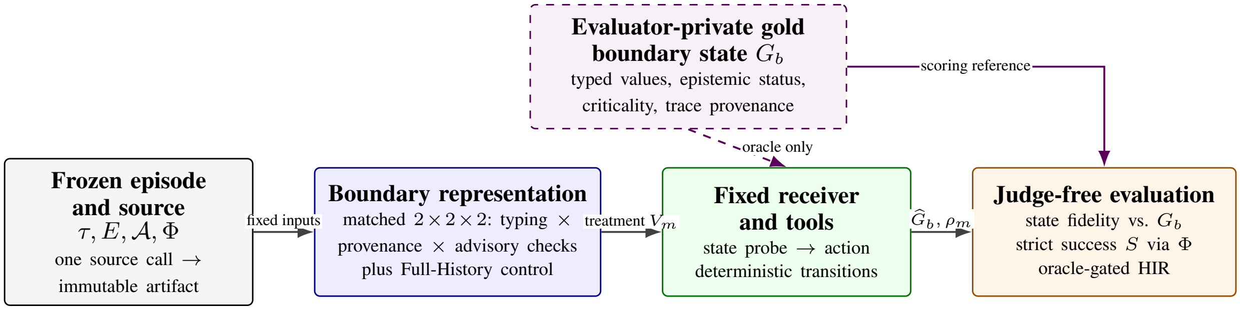

Figure 1: HandoffBench intervenes on the receiver-visible boundary representation while holding the authenticated trace, environment, receiver, tools, budget, and seed fixed. The gold state G_b is evaluator-private: it scores probe fidelity and is exposed to the same receiver only in the Gold Oracle arm. Joint oracle success and treatment failure define handoff-induced regression (HIR).

branch.

The evaluator proposal is a task-critical set of typed atomic claims,

$$G_b = \{g_i = (k_i, v_i, q_i, p_i, \ell_i, w_i)\}_{i=1}^{n}, \tag{1}$$

with canonical key k_i, normalized value v_i, epistemic status $q_i \in \{\textsc{known}, \textsc{unknown}, \textsc{contradicted}, \textsc{n/a}\}$, authenticated provenance p_i, criticality ℓ_i, and prespecified weight $w_i > 0$. A proposed claim is retained only when its omission or mutation can change a legal next action or Φ; open status is itself state and is never imputed. Executable preconditions remain in action rules, and tool-execution metadata is scored through workflow events, so neither is counted again in the primary semantic-state score. Candidate families must be independently double-annotated and adjudicated before sealing; model outputs are unavailable during this process.

Judge-free evaluator. Every receiver first emits the same schema-constrained state probe and then one schema-constrained action. Values are compared by declared canonicalizers (e.g., ISO dates, normalized identifiers, and unordered sets), never by an LLM judge. Within each $(\text{category}, k)$ group, gold and predicted claims are assigned one-to-one by deterministic maximum-weight bipartite matching. Hence a duplicated prediction can receive credit at most once and all remaining copies are false positives; duplicated gold atoms likewise require distinct predictions. For a matched pair $(g_i, \hat{g}_j)$, let r_{ij} be overlap measured relative to the gold value and p_{ij} overlap measured relative to the prediction; both reduce to exact equality of status and normalized value for non-set claims, while set claims use directional element overlap. We report the resulting weighted micro diagnostics,

$$R_s = \frac{\sum_{(i,j)\in\mathcal{M}} w_i r_{ij}}{\sum_i w_i}, \qquad P_s = \frac{\sum_{(i,j)\in\mathcal{M}} \bar{w}_{c(j)} p_{ij}}{\sum_j \bar{w}_{c(j)}}, \tag{2}$$

where $\bar{w}_c$ is a fixed category weight rather than a model-supplied weight. The assignment optimizes the combined directional recall and precision credit. Distinct from these weighted micro diagnostics, headline transfer fidelity is the unweighted mean of per-field F_1 over populated primary semantic fields, using the same field-local one-to-one matching. We additionally report category recall, contradiction among predictions sharing a gold key, unsupported-key rate, and provenance validity.

The simulator checks action arguments, ordering, call limits, public preconditions, and role allow-lists before applying deterministic user or tool transitions. Strict success is $S(e, m) = \mathbf{1}[\Phi(E_{\text{terminal}}, \rho_m)]$. Episode-level safety and efficiency errors include action without scoped consent, unsupported commitment reports, missed preconditions, duplicate stable tool calls, unnecessary re-asks, and role violations. A gold-state oracle uses the same receiver and environment but exposes canonical G_b. We count a paired handoff-induced regression only when the oracle succeeds:

$$\text{HIR}(e, m) = \mathbf{1}\{S(e, o) = 1 \;\wedge\; S(e, m) = 0\}. \tag{3}$$

Its reported rate is the mean of this joint indicator over all scheduled pairs, not failure probability conditional on oracle success. This avoids labeling failures that persist under perfect transfer as handoff-caused.

Worked episode. In sealed task repl_v3_procurement_003, authenticated events state that the requested target is option A, scope is B, authorization is unknown, and prerequisites are ready. The gold interface contains the corresponding typed atoms, including `requested_target_id=A` linked to the user event and `authorization_state=(unknown,null)` linked to policy. The factorial view renders one generated artifact either as fixed-field prose or typed atoms; provenance adds trace IDs, and checks add computed public-predicate statuses (e.g., `authorization_state=unknown: satisfied`). In the typed, trace-provenance, checks-on advisory cell for Qwen seed 202, the artifact omitted target A, the receiver selected option B, and the episode failed. In its matched Gold Oracle run, the same receiver observed A and completed the legal authorize–purchase sequence. This case illustrates the scored path from trace to boundary state to representation to action, not an additional statistical result.

Representation Factorial

The mechanism-identification experiment is a representation-only $2 \times 2 \times 2$ factorial: typing (fixed-envelope free-form versus typed atoms) $\times$ provenance (absent versus trace-linked) $\times$ checks (absent versus executable). All eight cells expose the same ten semantic fields. For each $(e, \text{model}, \text{seed})$ block, the source receives the identical frozen trace, atomic evidence, public predicates, prompt, schema, and token budget. Crucially, the source is called once per block: its immutable generated artifact, raw output, prompt/schema hashes, and logical usage record are reused across all eight cells. A fairness audit rejects an incomplete cube or any block with unequal source output, prompt, schema, or usage hashes.

The factors alter serialization, not source information. Typing renders the same atoms either as $\{k, q, v\}$ records or as deterministic prose within the fixed field envelope. Provenance-off cells omit bindings but do not withhold evidence from the source. Checks-off cells omit evaluated statuses but do not withhold public predicates. In checks-on cells, the framework evaluates those predicates against the generated source state and serializes each condition with `satisfied`, `missing`, or `contradicted` status for the receiver; it does not introduce evaluator gold. Every raw artifact is retained unchanged. Validation produces a sidecar and, where relevant, visible error annotations; it neither supplies corrected values nor quarantines a claim in the primary experiment.

All factorial cells use advisory execution. Executable checks therefore mean "represent a computed status," not "block an action." Action enforcement is an orthogonal binary intervention: the secondary enforced arm holds the all-on representation fixed and reports prevented violations, false blocks, and success. It is not pooled into factorial effects or described as a representation gain. Legacy Full History, Free Summary, Structured Payload, all-on Capsule, and Gold Oracle remain descriptive controls outside the factorial.

Confirmatory Statistics

Before any confirmatory call, we content-hashed and sealed a project-local analysis plan, hypotheses, multiplicity family, population, and execution configuration. We call this a pre-execution preregistration, but it was not deposited in a third-party registry. The population contains 200 independently authored families (40 in each of five domains), sealed after annotation, adjudication, replacement, and overlap audit. We evaluated Qwen2.5-14B-Instruct and Ministral-3-14B-Instruct-2512 with seeds 101 and 202, temperature .7, at most 1,600 output tokens and four receiver turns. Model and seed rows are repeated measures nested within task family, giving $200 \times 2 \times 2 \times 11 = 8{,}800$ scheduled cells. Analysis is intent-to-treat (ITT): every scheduled task–model–seed–condition cell remains in its assigned condition. Provider, schema, or parse failures receive zero strict success and state credit, remain in all denominators, and retain their raw calls and costs.

For the balanced factorial, factors are effect-coded $x_f \in \{-1, +1\}$. We estimated all main effects, all two-way interactions, and the three-way interaction; for term T, the balanced contrast is $2\,\mathrm{E}[Y \prod_{f \in T} x_f]$. Strict success is primary; state F_1, critical error, HIR, tokens, calls, and validator cost are secondary. Confidence intervals use percentile bootstrap resampling of independent task families, carrying along all model, seed, and condition rows. Binary paired contrasts use the same family-cluster bootstrap; exact McNemar tests are sensitivity analyses because repeated model/seed observations are clustered. Optional post-confirmatory mixed diagnostic models may include family and model intercepts, but they are not part of the confirmatory evidence or multiplicity family and are reported only with convergence diagnostics and any fallbacks.

The confirmatory multiplicity family contains (i) Structured versus Gold Oracle strict success and (ii) the advisory-check factorial main effect on strict success. Their two-sided p-values are Holm-adjusted; the directional checks hypothesis is supported only if its adjusted two-sided interval excludes zero in the positive direction. Typing, provenance, interactions, legacy contrasts, and domain/model subgroups are two-sided secondary or exploratory analyses and cannot be promoted after inspecting test outcomes. Development-only quantities, if shown, are labeled exploratory and reported separately.

The final analysis combines the intact 4,400-row Ministral arm from the sealed v3.3 execution with a fresh, sealed 4,400-row Qwen v3.4.1 full-arm rerun. An externally documented service termination invalidated the entire earlier Qwen arm; none of its rows were reused. The replacement was sealed before rerun calls, used a fresh output root, and recorded zero resumed and 4,400 newly written rows.

3 Benchmark Construction

Unit of construction. HandoffBench targets state transfer at a fixed control boundary rather than routing quality. Each episode fixes an authenticated upstream trace, source and receiver roles, the boundary cut, a public action contract, deterministic mock tools and user replies, and a machine-executable terminal predicate. The evaluator-side boundary state is a set of typed atomic claims with exact trace provenance and epistemic status (known, unknown, contradicted, or not applicable). We include a claim only when changing or removing it can alter a legal next action or terminal success. Goals, user constraints, verified business facts, ordinary unresolved slots, scoped consent, and organizational policy checks are mutually exclusive roles. Executable action preconditions remain in the public contract, and tool-execution metadata remains in the trace; neither is duplicated as a semantic claim for primary state F_1.

Generation and validation. We first authored workflow blueprints—roles, causal trace, authority boundary, plausible alternatives, legal sequence, and first user-impacting action—then instantiated synthetic dates, monetary values, statuses, scopes, attributes, and opaque identifiers with deterministic generators. Families are the split unit: entity replacement or paraphrase does not create a new family. The original pool contained 200 families evenly divided among travel, commerce, procurement, enterprise IT, and scheduling.

Every label-free packet was independently annotated by two isolated LLM-agent roles and locked before comparison. On 990 claims, claim F_1 was .961 and exact action-sequence agreement was 174/200; category and criticality agreement had Cohen's $\kappa = .945$ and $.242$, respectively, while status, typed value, and provenance agreed on 990/990 claims. Disagreement-only adjudication covered all 739 queued entries and rejected 24 tasks. Separately, an independent review agent blind to labels and adjudication examined 39 agreement-only tasks flagged by static validity risks and rejected all: 34 had ambiguous terminal actions, four had ambiguous sequence order, and one lacked public terminal semantics.

We generated 63 new families to replace these disjoint exclusions, without inspecting any candidate-model output, and double-annotated them anew. The two annotation agents agreed on all 63 action sequences and on identity, category, status, value, and provenance for all 299 claims. Criticality agreement was 252/299 ($\kappa = .693$); a separate adjudication agent resolved all 47 criticality disagreements, rejecting no replacement. The final population is therefore 137 retained originals plus 63 replacements, again 40 per domain (Table 1).

Isolation and lineage. The two annotation-agent roles received independently shuffled packets containing only the trace cut, role boundary, public action schemas, and deterministic scripted transitions. They did not receive source annotations, evaluator predicates, development outputs, family blueprints, or one another's responses. Each response was schema-validated and content-hashed before comparison. The adjudicator saw only queued field or sequence disagreements; exact-agreement items were not reopened. Likewise, the independent validity-review agent saw only the 39 designated public packets, not A/B labels, queues, adjudication decisions, or evaluator gold. This separation matters because agent agreement alone cannot show that a public contract identifies a unique terminal behavior.

Rejected originals remain in the audit lineage but are absent from the final task files. All 63 replacements use new task and family identities and were subjected to the same schema, provenance, grounding, mutation, and simulator checks before inclusion. Sixteen replacements had exact agreement and required no adjudication; the other 47 each contributed a criticality disagreement that was resolved without changing public task evidence. A composite readiness artifact binds original and replacement locks, agreement reports, adjudications, the blind review, final task files, and final audit by SHA-256. This makes the reported 200-task composition reproducible without exposing model outcomes, because none existed during selection.

Final static audit and seal. All prespecified hard checks pass: every final task has one public legal terminal sequence without access to the hidden predicate, grounded irreversible arguments, valid provenance and impact labels, and successful deterministic oracle execution. There are no development collisions in family ID, entity pool, seed, exact or normalized trace, or exact action graph, and no trace bigram pair reaches .80. The split is bound to seal `hb-v3.1-9671bf1ff2d5-20260721` and canonical dataset hash `9671bf1ff2d5...d892a93`. No candidate-model call occurred during construction, annotation, replacement, auditing, or sealing.

Table 1: Agent annotation, sealing, and static diagnostics for the final v3 split. Probe rates are not model-performance results.

Statistic	Count / rate
Sealed families (five domains)	200 (40 each)
Retained originals / replacements	137 / 63
Original adjudication / blind-review rejects	24 / 39
Replacement disagreements / rejects	47 / 0
Unique public legal terminal sequence	200 / 200
Exact-copy oracle execution	200 / 200
Catalog-only / predicate-only success	7.0% / 6.0%
Oracle sequence + enum-first success	61 / 200 (30.5%)
Exact development identity/trace collisions	0 / 0
Normalized topology flags vs. development	69
Lexical pairs at Jaccard $\geq .80$	0
Candidate-model calls before sealing	0

The leakage audit obtains 14/200 from a catalog-only heuristic and 12/200 from predicate-only features. A stronger diagnostic, accurately called *oracle sequence plus enum-first*, is given the evaluator-side ordered action-name sequence and guesses the first public enum value; it succeeds on 61/200 (30.5%; Wilson 95% CI [24.5, 37.2]). It therefore does not measure what a receiver can infer from public action names alone. Exact copy succeeds on 200/200 and establishes simulator consistency only.

Sixty-nine final tasks share a normalized action-graph hash with development tasks. The normalization erases action names, keys, and values, and can collapse common ask/authorize/commit skeletons; this is a topology diagnostic rather than evidence of semantic duplication. Conversely, its concentration means that we do not claim a complete structural holdout. These diagnostics were reported without post-hoc filtering. The dataset remained byte-identical throughout confirmatory execution. The final v3.4.1 execution seal retains the v3.1 dataset hash and binds the model snapshots, prompts, generation settings, and analysis contract.

4 Closest Related Work

Learned communication studies optimize messages for cooperative tasks (Sukhbaatar, Szlam, and Fergus 2016; Foerster et al. 2016), while stateful tool-use and collaboration benchmarks measure policy compliance, terminal success, and team-level coordination (Yao et al. 2024; Lu et al. 2025; Zhu et al. 2025). These lines do not expose a scored semantic state at a fixed, manipulated role boundary. The broad idea of evaluating agent handoffs is also not new. Most directly, *Handoff Debt* interrupts coding agents at deterministic repository checkpoints and compares repository-only, raw-trace, summary-note, and structured-note views, measuring task resolution and successor rediscovery cost (KC and Budathoki 2026). We therefore do not claim the first handoff benchmark, controlled takeover, or structured transfer artifact. Our estimand is instead the fidelity of a semantic interface at an inter-role boundary: each episode exposes an evaluator-only set of atomic claims with epistemic status and trace evidence, and

Work	Primary object	Distinction of HandoffBench
Handoff Debt	Coding takeover across four context views	Atomic gold boundary state with epistemic and authority claims; oracle-gated regression rather than resumability.
NeuroState-Bench	Within-agent commitment probes	A manipulated predecessor–successor boundary and trace-grounded claims.
Provenance-Guard	Evidence-based action screening	Provenance is a representation factor; action blocking is a separate intervention.

Table 2: Claim-level positioning against the three closest mechanism-level collisions. Structured handoffs, state probes, and provenance-aware blocking are prior art; our target is controlled measurement of semantic state transfer across a role handoff. Concurrent benchmark comparisons appear in the text.

tests which claim errors precede policy-invalid or regressive actions. This also differs from state-probe and runtime-guard work (Jia 2026; She, Liang, and Kang 2026) (Table 2). Concurrent work makes the collision sharper. PACT treats inter-agent messages as compact public action-state updates and evaluates accuracy–cost trade-offs (Huang, Wu, and Zhang 2026); AgentAsk annotates edge-level data gaps, referential drift, signal corruption, and capability gaps, then learns when to clarify (Lin et al. 2026); and EntCollabBench objectively scores role-specialized, permission-isolated enterprise workflows in which context transfer is one bottleneck (Yu et al. 2026). HandoffBench does not claim these messaging, repair, or enterprise settings. It contributes evaluator-visible atomic boundary state and matched representation interventions that estimate oracle-gated boundary regression and test serialization mechanisms under fixed evidence.

5 Development Findings and Design Adaptation

We used six counterfactual travel families only to debug the protocol and set the confirmatory design; these data are excluded from all confirmatory tables. Across Qwen2.5-14B and Ministral-3-14B (48 scheduled episodes per legacy view), strict success was 68.8% for the all-on executable handoff capsule (EHC), 75.0% for the schema-matched Structured payload, 47.9% for Free Summary, 91.7% for Full History, and 93.8% for the Gold Oracle. The EHC–Structured difference was -6.25 percentage points (family-clustered 95% CI $[-22.9, 12.5]$). This is a negative development result: it does not support generic EHC superiority, and we retain rather than tune it away.

A representation-matched $2 \times 2 \times 2$ development factorial (typing, trace-linked provenance, and advisory checks) scheduled 384 cells for two model families; 381 completed provider/parsing successfully, and all cells remain in the intent-to-treat denominator. Strict-success main effects were $+2.60$ pp for typing (95% CI $[0.52, 4.69]$), -6.77 pp for provenance ($[-15.63, 2.60]$), and $+6.77$ pp for advisory checks ($[3.65, 9.38]$). The provenance–checks interaction was -4.69 pp ($[-10.42, -0.52]$); provenance also increased mean critical-error count by 0.042 ($[0.010, 0.078]$). With only six independent families, these estimates are diagnostic, not population evidence. They changed the study's thesis from "all-on capsules are superior" to two falsifiable benchmark questions: whether boundary regression exists, and which representation mechanisms help or harm under matched information.

Power was likewise treated as a pre-test design input. A fixed-seed simulation with model repeats nested within family and family-cluster bootstrap inference estimated, at ICC = .10, power of .441 for an 8-pp effect and .633 for a 10-pp effect at 120 families; at 200 families the corresponding estimates were .691 and .847. We therefore expanded the candidate pool to 200 independently authored families (40 per domain) before annotation or any candidate model call. These simulated values are assumptions, not observed test effects, and do not permit optional stopping.

6 Confirmatory Results

The sealed matrix contains all 8,800 scheduled ITT cells: 200 independent families, two models, two seeds, and eleven conditions. The only two confirmatory tests sealed before execution use family-cluster percentile-bootstrap intervals (10,000 draws) and Holm-adjusted two-sided p-values. All other factorial, endpoint, condition, model, and domain estimates below are secondary or exploratory. Across the matrix, 178 cells (2.02%; 97 Ministral and 81 Qwen) had non-OK model-output, schema, or parse status; the prespecified ITT rule retained them with zero success and state credit. Their processing points were nine source-transfer parses, eight receiver-output parses, 86 receiver-state schema/semantic checks, and 75 receiver action-catalog checks. The respective Qwen/Ministral splits were 0/9, 1/7, 49/37, and 31/44. Non-OK rows occurred in every non-oracle condition rather than a single treatment; Gold Oracle had zero. Complete condition-by-model counts and the raw failure taxonomy are included in the released non-OK audit.

Test	Rates (%)	Δ pp [family 95% CI]	p_{Holm}
Structured / Oracle	71.63 / 91.13	-19.50 $[-23.63, -15.38]$	$< .001$
Checks off / on	70.19 / 73.81	$+3.63$ $[+1.91, +5.44]$	$< .001$

Table 3: The two prespecified confirmatory strict-success tests. Rate order matches the test labels; effects are percentage points. All ITT rows are retained and inference is clustered by task family.

Both prespecified tests rejected their null hypotheses after Holm correction. Structured transfer reached 71.6% strict success, compared with 91.1% under the Gold Oracle, for a paired effect of -19.50 percentage points (95% CI $[-23.63, -15.38]$; $p_{\mathrm{Holm}} < .001$). Its oracle-gated HIR incidence over all scheduled pairs was 21.75% (95% CI $[17.88, 25.63]$): in these cells, the oracle succeeded while Structured failed. In the matched representation factorial, advisory executable checks improved strict success by 3.63 points on average (95% CI $[1.91, 5.44]$; $p_{\mathrm{Holm}} < .001$), supporting the prespecified directional hypothesis.

Secondary, unadjusted strict-success estimates were $+1.44$ points for typing (nominal 95% CI $[0.06, 2.84]$) and -1.63 points for trace-linked provenance ($[-2.81, -0.47]$). All two-way interaction intervals crossed zero, as did the three-way interaction. These mechanism diagnostics are outside the confirmatory multiplicity family and do not establish general typing benefits or provenance harms. Secondary endpoints were directionally consistent with a small diagnostic value for typing: macro state F_1 rose by .0098 and mean critical errors fell by .0766. Checks reduced calls by .075, input tokens by 129, and output tokens by 37 on average, while provenance increased input use by 891 tokens. These endpoint intervals are likewise unadjusted.

An explicitly post-confirmatory, unadjusted model diagnostic found a checks effect of $+7.06$ points for Ministral (family-bootstrap 95% CI $[3.81, 10.50]$) but $+0.19$ for Qwen ($[-0.94, 1.38]$); the Qwen-minus-Ministral difference was -6.88 points ($[-10.44, -3.44]$). Thus, the pooled gain did not replicate in Qwen. These exploratory intervals receive no multiplicity correction and do not support a universal model or domain claim. Descriptively, Full History achieved 90.4%, close to the Oracle's 91.1%, despite much lower normalized first-probe state F_1; successful action selection need not require explicit recovery of every probed atom. The all-on advisory and enforced arms achieved 73.1% and 74.1%, respectively, but this difference is not a confirmatory enforcement test.

Execution and reproducibility. The analysis binds dataset seal `hb-v3.1-9671bf1ff2d5-20260721` to execution seal `hb-v3.4.1-exec-9671bf1ff2d5-20260721`. It combines 4,400 retained Ministral rows with 4,400 fresh Qwen rows; the invalidated earlier Qwen arm contributes zero rows. The Qwen ledger records 4,400 written and zero resumed rows, and the infrastructure audit found no provider infrastructure error. Bootstrap seed 2027, analysis code hashes, all raw-input hashes, and generated artifact hashes are recorded in the released provenance manifest.

7 Ethics and Limitations

All scenarios, entities, traces, policies, and tool outputs are project-authored synthetic data, not customer records or private company policies; no real person is intentionally represented. Benchmark annotation used two isolated LLM-agent roles plus a disagreement-only adjudication agent, not recruited human annotators. Their packets, locked responses, and hashes are retained, but the exact underlying annotation-model/runtime identity was not recorded; agreement therefore measures replicated agent judgments rather than human validity. An independent human audit remains necessary before claiming human-grounded gold labels. Generative-AI systems also assisted code generation, literature triage, experiment orchestration, manuscript drafting, and internal review. The human authors remain responsible for all submitted material; agent-produced citations and claims require author verification before submission. Future free-text annotations could introduce personal information, so exports require PII/secret review. The tasks simulate consequential purchases, refunds, identity and access checks, accessibility requests, and travel-document workflows. Their simplified policies are unsuitable for operational, legal, medical, financial, employment, or other decisions about real people.

The sealed split is synthetic and may contain lexical regularities, simplified jurisdictional assumptions, and limited real-world coverage. Its five authored domains, deterministic user/tool simulators, and single handoff per episode do not reproduce naturally occurring organizational communication, routing errors, long-horizon memory, or emergent multi-party coordination. Isolated agent reconstruction and deterministic checks establish procedural consistency, not human or ecological validity. The 30.5% oracle-sequence-plus-enum-first diagnostic reveals ordered-enum regularity under an evaluator-provided workflow path; it is not name-only performance. Moreover, 69 normalized-topology overlaps with development tasks preclude a claim of complete structural holdout, although there are no exact development identity, trace, or action-graph collisions. A post-seal replication can adapt separately licensed ToolSandbox scenarios and the telecom subset of τ^2-Bench (Lu et al. 2025; Barres et al. 2025), but must report each source separately and may not replace rejected primary families after outcomes are observed. The confirmatory study is limited to two local 14B open-weight instruction models, two seeds, synthetic single-handoff tasks, and deterministic mock tools; it does not establish behavior in long-running collaboration or real organizations. The Gold Oracle is an identifying intervention, not a deployable baseline. The pooled checks result averages eight representation cells and does not imply that enforcement, every validator, model, or domain benefits. Typing, provenance, secondary endpoints, and subgroup intervals were outside the confirmatory multiplicity family. The retained Ministral and replacement Qwen arms were executed at different times, so an unmeasured environment batch effect remains possible despite fixed snapshots, configuration, and data. A post-run hardware/package snapshot is available for the replacement Qwen arm, but no equivalent package-version snapshot was captured for the retained Ministral arm. We disclose the external service termination that invalidated the earlier Qwen arm; the decision used infrastructure logs rather than scientific outcomes, and all 4,400 earlier Qwen rows were excluded.

Project-authored code is released under Apache-2.0 and project-authored synthetic data under CC BY 4.0; the manuscript, author kit, model weights, and third-party materials are excluded from those grants. The sealed tasks cannot be repaired or filtered after model outcomes are observed, and the reported confirmatory effects are not representativeness claims.

8 Conclusion

Reliable multi-agent systems require more than correct routing. Treating the handoff boundary as a measurable interface exposed a 19.5-point loss from Structured transfer relative to the Gold Oracle and 21.8% oracle-gated HIR incidence over all scheduled pairs. Advisory checks recovered 3.6 points on average but did not close the boundary gap, and an exploratory split found the gain only in Ministral, not

Qwen. Secondary estimates suggest that serialization mechanisms need not combine monotonically: typing was mildly beneficial, while the tested trace-linked provenance implementation incurred a small negative success estimate and a large token cost. These findings support explicit boundary-state evaluation and targeted validation, not a blanket claim that richer capsules or enforcement are universally superior. HandoffBench's contribution is therefore not a universal capsule recipe, but an outcome-linked, judge-free protocol for making the handoff interface experimentally measurable.

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